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*Biological Chemistry* ’Just Accepted’ paper
ISSN (online) 1437-4315
DOI: 10.1515/hsz-2017-0171

**Review**

# Eremophilane-type sesquiterpenes from fungi and their medicinal potential

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## Abstract

Eremophilanes are sesquiterpenes with a rearranged carbon skeleton formed both by plants and fungi, however, almost no plant eremophilanes are found in fungi. These eremophilanes possess mainly phytotoxic, antimicrobial, anticancer and immunomodulatory properties and in this review fungal eremophilanes with bioactivities of potential medicinal applications are reviewed and discussed. A special focus is set on natural products bearing highly functionalized fatty acids at C-1 or C-3 position of the eremophilane backbone. Many of these fatty acids seem to contribute to the bioactivity of the metabolites enhancing the activity of the sesquiterpene moieties. Several approaches for optimization of these natural products for clinical needs and testing of the resulting derivatives are presented and discussed. The combination of identification of bioactive natural products with their subsequent improvement using a variety of genetical or chemical tools and the pharmacokinetic assessment of the products is here presented as a promising approach to new drugs.

**Keywords:** antibiotic; cytotoxin; eremophilane; fungi; immunomodulator; sesquiterpene

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## Introduction

Terpenes constitute one of the largest families of natural compounds. They consist of isoprene units of the sum formula $nC_5H_8$. These isoprenes are attached head-to-tail to each other, a principle known as the Isoprene Rule, which was first postulated by Otto Wallach in 1887 (Wallach, 1887). Sesquiterpenes, consisting of three isoprenes, are probably the most diverse group of terpenes. In 1932, three of these sesquiterpenes have been isolated from *Eremophila mitchelli* and named eremophilones (Bradfield *et al.*, 1932). It took seven years to realize that the underlying hydrocarbon, eremophilane **1-1**, did not follow the Isoprene Rule (Penfold and Simonsen, 1939). In the following years more and more natural products possessing the eremophilane skeleton, have been isolated from plants with a focus on species of the Asteraceae family (Bohlmann *et al.*, 1977; Pinder, 1977). A recent review on the occurrence and biological activities of eremophilane sesquiterpenes focused entirely on natural products from plants (Hou *et al.*, 2014). However eremophilanes are also synthesized by fungi, but it took more than four decades after the discovery of eremophilanes in plants before phomenone **1-2**, the first eremophilane from fungi, has been identified (Riche *et al.*, 1974).

Today about 150 different eremophilanes have been reported from various fungi (Ebel, 2010) but only two of these compounds have been detected both in fungi and plants. These metabolites are petasol **1-4** from a *Penicillium* sp. (Lin *et al.*, 2014) and from the plant *Petasites* spp. (Neuenschwander *et al.*, 1979) and mairetolide F from *Xylaria* sp. (Isaka *et al.*, 2010) and from the plant *Senecio mairetianus* (Pérez-Castorena *et al.*, 2006). This fact underlines the uniqueness of fungal eremophilanes, which is further underlined by the observation that most eremophilane moieties from fungal metabolites are enantiomeric to the corresponding sesquiterpene part in plants. A unique feature of several fungal eremophilanes are unusual carboxylic acids at C-1 or C-3. Because these eremophilanes are so unusual they have been summarized in Figures 4 and 5. Currently, a focus for the search of novel fungal sesquiterpenes seems to be in marine fungi (Wang *et al.*, 2015) and in the genus *Penicillium* (Nicoletti and Trincone, 2016).

Probably all eremophilanes have their specific bioactivities and ecological functions otherwise the producing fungus would not waste energy and resources for their production. However, only for about half of the known fungal eremophilanes bioactivities have been identified and reported in the literature. In this review, only eremophilanes from fungi with reported bioactivities will be presented and their biological activities discussed. The majority


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of reported activities are those of phytotoxins, antimicrobials, protein inhibitors, immunomodulators and cytotoxins. Where possible the focus will be on the targets for the activities of fungal eremophilanes together with details on their structure-activity relationships. Finally, an outlook on the future of the applicability of fungal eremophilanes and their chemical derivatives will be given.

## Biosynthesis of eremophilanes

Biosynthesis of the eremophilane skeleton has been elucidated mainly by the application of stable isotopes and NMR spectroscopy. The synthesis follows the standard mevalonate pathway and involves farnesylpyrophosphate. Cyclases form then the bicyclic eudesmane skeleton via a germacrene A intermediate. The cyclase aristolochene synthase catalyzes the elimination of a proton from C-12 and the addition of another proton anti to the isopropylidene side chain of germacrene (Miller *et al.*, 2008). In the resulting eudesmane precursor, the C-15 methyl group is then shifted from C-10 to C-5 leading to the eremophilane skeleton (Figure 1). By feeding [1,2-$^{13}$C]-acetate and determining the $^{13}$C-$^{13}$C couplings of the formed PR-toxin **1-5** the shift of the C-15 methyl group could be demonstrated (Moreau *et al.*, 1981). Administration of [4-$^{2}H_2$]mevalonic acid and $^{2}$H NMR spectroscopy of the resulting eremophilane showed that the migration of the methyl group is accompanied by a 1,2-hydride shift from C-5 to C-4 (Gorst-Almann and Steyn, 1982). It should be noted here that the cyclase of *Fusarium fujikuroi* forms eremophilene, which is enantiomeric to the one from plants (Burkhardt *et al.*, 2016), but has the same absolute configuration as the one from the bacterium *Sorangium cellulosum* (Schifrin *et al.*, 2015). After formation of the eremophilane skeleton, the intermediate is further transformed via eremofortin B (Moreau *et al.*, 1980) and other intermediates to the final product PR-toxin (Riclea and Dickschat, 2015). As more and more genome sequences are becoming available (Garcia-Estrada and Martin, 2016) more cyclases forming eremophilanes will be discovered and comparisons of gene organizations in different organisms will lead to the improved formation of bioactive eremophilanes for biotechnological applications.


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## Antimicrobial and antagonistic eremophilanes from fungi

Xylarenone A **1-6** and its acetate xylarenone B **1-7**, have been isolated from the endophytic fungus *Xylaria* sp. NCY2, which was obtained from *Torreya jackii*. Antibacterial activities against Gram positive bacteria was demonstrated and here the acetylated xylarenone B was slightly more active than its alcohol xylarenone A. They also exhibited moderate antitumor activities against HeLa cells but were not active against yeasts (Hu *et al.*, 2008). Xylarenone A together with other xylarenones was also found in an unidentified endophytic fungus isolated from *Alibertia macrophylla* (Rubiaceae) and tested for their ability to inhibit proteases. While both xylarenone A **1-6** and C **4-55** were inhibitors of pepsin $IC_{50}$ = 2.43 μM and $IC_{50}$ = 0.288 μM, respectively), xylarenon C could also inhibit subtilisin with $IC_{50}$ = 0.462 μM (Oliveira *et al.*, 2011). Phomadecalins A-D were isolated from *Phoma* sp. NRRL 25697, a fungus colonizing the stromata of a *Hypoxylon* species. Phomadecalins A **1-8**, B **1-9** and D **1-10** showed moderate activities against *Bacillus subtilis* ATCC 6051 and *Staphylococcus aureus* ATCC 29213. Similar to the xylarenones, no activities were found against yeasts (Che *et al.*, 2002). Another endophytic fungus, *Microdiplodia* sp. KS 75–1, also produced phomadecalin C and D together with dihydro-xylarenone A **1-11** and the acetate **1-12**. In this work, the stereochemistry of phomadecalin C and D has been corrected and it was found that dihydro-xylarenone A and **1-12** were active against *Pseudomonas aeruginosa* (Hatakeyama *et al.*, 2010). From the fermentations of a *Dacrymyces* sp., the antibiotic dacrymenone **1-13** was isolated which showed weak activities against the bacterium *Micrococcus luteus* and several yeasts (Mierau *et al.*, 2003). A *Microsphaeropsis* sp. was isolated from the marine sponge *Myxilla incrustans* and found to produce several metabolites. One of them was the eremophilane microsphaeropsisin **1-14** with had weak antibacterial activity and inhibited the fungus *Eurotium repens* (Höller *et al.*, 1999) (Figure 2).

The soil fungus *Penicillium copticola* PSURSPG138 produced several metabolites, among them penicilleremophilane A **1-15** with the unusual 3-chloro-4-hydroxyphenylacetyl ester and sporogen AO-1 **1-3**. These two compounds belong to the very small group of eremophilanes possessing antimalarial activities. Tests against *Plasmodium falciparum* revealed for sporogen AO-1 significant antimalarial activity with $IC_{50}$ of 1.53 μM compared to the cytotoxic activity against Vero cell line of $IC_{50}$ of 4.23 μM. Penicilleremophilane A was with 3.45 μM against *P. falciparum* about half as active but had a much weaker cytotoxic activity than sporogen AO-1 (Daengrot *et al.*, 2015). From the unidentified xylariaceous endophytic fungus YUA-


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026 two eremophilane sesquiterpenes, eremoxylarins A **4-51** and B **4-53**, bearing unusual esters were isolated. While these compounds were not active against fungi, they showed good activities both against *Staphylococcus aureus* and *Pseudomonas aeruginosa*. The minimum inhibition concentration (MIC) of eremoxylarin A against *Staphylococcus aureus* was 12.5 μg $ml^{-1}$ and 25 μg $ml^{-1}$ for eremoxylarin B; MIC against *Pseudomonas aeruginosa* were 6.25 μg $ml^{-1}$ and 12.5 μg $ml^{-1}$, respectively (Shiono and Murayama, 2005). These compounds differ only in the ester moieties and it is interesting to note that eremoxylarin B was only half as active as eremoxylarin A, which points to an influence of the methylated carboxylic side chains. However, in animal models it was shown that eremoxylarin A has considerable cytotoxicity (Efremenkova *et al.*, 2015). A related compound, 1-(xylarenoneA)xylariateA **4-59** was isolated from the endophytic *Xylaria* sp. NCY2 of *Torreya jackii* Chun. This eremophilane ester had similar antibiotic properties compared to the eremoxylarins (Hu *et al.* 2010).

Microorganims tend to escape antibiotics by developing resistance mechanisms. These antibiotic resistances pose severe problems in the clinic for the treatment of bacterial infections and more and more bacterial strains with resistances against several of the commercial antibiotics have been isolated from patients. One way to combat this tendency is the identification and development of compounds, which inhibit main resistance mechanisms. One widespread pathogen is *Staphylococcus aureus*, which causes as methicillin-resistant *S. aureus* (MRSA) severe problems in the clinic. Its main mechanism of resistance to aminoglycosides is their inactivation by enzymes. Here, the bifunctional enzyme AAC(6′)-Ie/APH(2″)-Ia catalyzing both phosphorylation at 2″-hydroxyl and acetylation at 6′-amino group of aminoglycosides plays a crucial role (Ferretti *et al.*, 1986) and specific inhibitors against this enzyme are urgently needed. *Paraphaeosphaeria* sp. TR-022 produced such compounds as berkleasmin A **4-60** and F **4-62,** which restore sensitivity of MRSA against aminoglycosides. Both, berkleasmin A and F, having identical ester moieties, display similar activities, which may indicate the dominant role of the unusual fatty acid in these metabolites. Surprisingly, the activity of both compounds does not target AAC(6′)-Ie/APH(2″)-Ia because no inhibition of this enzymes could be observed (Suga *et al.*, 2016).


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## Fungal eremophilanes as inhibitors

Some fungal eremophilanes have been found to inhibit important enzymes. One of the first compounds of this class was integric acid **4-54** with a long and branched fatty acid at C-1 isolated from a *Xylaria* species which interacts with the integrase of the Human Immunodeficiency virus (HIV)-1. This enzyme catalyzes the integration of the viral DNA into DNA of the host cell. Because this enzyme is very specific for HIV-1 it became a potential target for the search for anti-HIV antiviral agents. Integric acid inhibits 3'-end processing and strand transfer reactions, both with $IC_{50}$ of 10 μM (Singh *et al.*, 1999). In an attempt to determine the minimal pharmacophore of the inhibitor, several derivatives have been prepared and tested for their ability to inhibit HIV-1 integrase. The results indicated that certain modifications are tolerated but the ester group at C-1, the acid groups at C-4 and the vinyl aldehyde moiety at C-11 are required for strand transfer activity. Surprisingly, the side chain alone developed significant inhibitory activity. The observations point to different structure activity relationships for the 3’ processing and strand transfer reactions of the integrase (Singh *et al.*, 2000). Biosynthetic studies of integric acid in *Xylaria feejeensis* revealed that the methyl groups at the side chains are derived from methionine (Srisapoomi *et al.*, 2015). Another protein important for HIV-infection is the regulatory protein HIV-1 Tat, which is required for virus replication. A screen identified the eremophilanes sporogen AO-1 **1-3** und petasol **1-4** as moderate inhibitors with $IC_{50}$ of 15.8 and 46.2 μM, respectively. The closely related 6-dehydropetasol was inactive. The toxicity of both compounds was higher than 100 μM. However, the macrocyclic lactones monorden and monocillin IV were much more active with a similar low cytotoxicity (Jayasuriya *et al.*, 2005).

Fermentation of *Xylaria persicaria* afforded the two eremophilane derivatives, xylarenal A **4-63** and B **4-64** with medium-length but unbranched acyl side chains. Both are antagonists of neuropeptide Y, which functions as a neurotransmitter in the peripheral and central nervous system. It has been associated with several physiological functions, e.g. circadian rhythms, feeding, anxiety, and learning. In mammals neuropeptide Y binds to several receptors with different affinities, one of them being Y5, which is associated with food uptake. Xylarenal A and B interact specifically with the receptor Y5, but only with moderate affinities ($IC_{50}$ = 6.8 and 3.6 μM, respectively) (Smith *et al.*, 2002). Because of their high selectivity xylarenals A and B may be good starting points for the development of drug candidates for the treatment of obesity.


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## Anti-inflammatory eremophilanes from fungi

Six eremophilanes, nigriterpenes A-F, were isolated *Xylaria nigripes*, which has long been used in traditional Chinese medicine for the treatment of insomnia and depression. Nigriterpene C **2-16** was found to inhibit lipopolysaccharide-induced inducible nitric oxide synthase ($IC_{50}$ = 8 μM), cyclooxygenase-2 expression ($IC_{50}$ = 17 μM), and NO production ($IC_{50}$ = 22 μM) in murine brain microglial BV-2 cells without any significant cytotoxicity. The potential anti-inflammatory effects of nigriterpene C may provide a rationale for the traditional medical uses of *X. nigripes* for treating insomnia and depression (Chang *et al.*, 2017).

To activate silent gene clusters the marine-derived fungus *Aspergillus* sp. SCSIOW2 was cultivated with the histone deacylase inhibitor suberohydroxamic acid or the DNA methyltransferase inhibitor 5-azacytidine resulting in the formation of novel secondary metabolites and dihydrobipolaroxin **2-17**, dihydrobipolaroxin B **2-18**, dihydrobipolaroxin C **2-19**, and dihydrobipolaroxin D **2-20** were identified (Figure 3). All four compounds exhibited weak NO inhibitory activities without cytotoxic effects and dihydrobipolaroxin was the most active of these compounds (Wang *et al.*, 2016). Cuspidatol **2-21** was obtained from the culture broth of the endophytic fungus *Trichothecium roseum*, isolated from a leaf of the tree *Taxus cuspidata*. It was applied in a mixed lymphocyte reaction (MLR) assay (responder: C57BL16 mice, stimulator: Bal b/C mice pretreated with mitomycin C). The metabolite had an $IC_{50}$ of 11 μg $ml^{-1}$, while not showing cytotoxicity (>100 μg $ml^{-1}$) against several human cell lines (Kawamura *et al.*, 2000).

From the medicinal plant *Annona muricata* the fungus *Periconia* sp. F-31 was isolated and screened for the production of secondary metabolites. From its culture broth periconianone A and B (Zhang *et al.*, 2014) and C-K (Liu *et al.*, 2016) were identified. Periconianone A **2-22** has an unusual tricyclic carbon skeleton with an additional bond between C-12 and C-4. Periconianone A **2-22** and B **2-23** inhibited lipopolysaccharide-induced NO production in mouse microglia BV2 cells with $IC_{50}$ values of 0.15 and 0.38 μM, respectively. When tested for anti-inflammatory activity indirectly by suppressing LPS-induced NO production in BV2 cells periconianone D **2-24**, G **2-25**, and K **2-26** revealed inhibition rates of 10%, 18%, and 16%, respectively, at a concentration of 1.0 μM. This was comparable to equally concentrated curcumin with an inhibition rate of 13% (Liu *et al.*, 2016). In the similar assay guignarderemophilane A **2-27**, C **2-28**, and D **2-29**, obtained from *Guignardia mangiferae*


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growing on the plant *Gelsemium elegans*, had $IC_{50}$ values of 15, 6 and 4 μM ($IC_{50}$ curcumin 4 μM) (Liu *et al.*, 2015). The endophytic fungus *Camarops* sp. was isolated from *Alibertia macrophylla* and produced also several xylarenones. None of the compounds had cytotoxicities below 100 μM. Xylarenones C, D, F and G were potent inhibitors of reactive oxygen species produced by stimulated neutrophils with inhibitory concentrations of xylarenone C **4-55** ($IC_{50}$ = 6.1 μM), D **4-56** ($IC_{50}$ = 5.7 μM), F **4-57** ($IC_{50}$ = 5.9 μM), and G **4-58** ($IC_{50}$ = 4.2 μM). As none of these compounds acted as scavengers of superoxide anions the effect was probably caused by NADPH oxidase inhibition (Gubiani *et al.*, 2014). Acremeremophilanes B - F **5-67** – **5-71** from *Acremonium* sp., isolated from deep-sea sediments, are unusual by bearing *trans*-4'-hexenoic acid at C-3. The cytotoxicity of these compounds was low with a survival rate of RAW 264.7 cells of >90% at 50 μM. Acremeremophilanes B - F showed inhibitory effects against NO production in LPS-activated RAW 264.7 macrophages with $IC_{50}$ values between 8-45 μM. Acremeremophilanes B **5-67** and E **5-70** were here the most active compounds (Cheng *et al.*, 2016).

## Cytotoxic fungal eremophilanes

A strain of *Periconia byssoides*, isolated from the sea hare *Aplysia kurodai*, produced a number of eremophilanes, the peribysins. In 2004 the isolation and structure elucidation of peribysin A - D **3-30** – **3-32, 3-34** was reported (Yamada *et al.*, 2004), two years later the structures of peribysin C **3-32** and D **3-34** have been revised (Koshino *et al.*, 2006) (Figure 4). Later on the same strain yielded peribysin I, H, E - G **3-33**, **3-35** – **3-38** (Yamada *et al.*, 2005, 2006). The absolute stereochemistry of peribysin G was determined by x-ray crystallography and CD spectra. Nevertheless, the enantiomeric pure synthesis of (+)-peribysin E yielded the enantiomeric configuration compared to peribysin G (Angeles *et al.*, 2007). It seems to be that the same strain produced both eremophilane enantiomers. These compounds are cell-adhesion inhibitors, inhibiting the adhesion of human-leukemia HL-60 cells to human-umbilical-vein endothelial cells (HUVEC). The mechanism of action is probably via inhibition of protein tyrosine kinase (Weber *et al.*, 1995). The inhibition by peribysins is more potent than the well-established inhibitor herbimycin A (Okabe *et al.*, 1994; Yamada *et al.*, 2005).

From an Antarctic deep-sea sample, *Penicillium* sp. PR19 N-1 was obtained which produced a number of eremophilanes of which compounds **3-39** and **3-40** displayed modest cytotoxicity against HL-60 cells with $IC_{50}$ values of 46 and 28 μM and against A-549 cells, with $IC_{50}$


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values of 83 and 5 μM respectively (Lin *et al.*, 2014). The chlorinated trinor-eremophilane **3-41** formed by the same strain was also active, showing $IC_{50}$ of 12 μM both against HL-60 and A549 cell lines (Wu *et al.*, 2013). Of the eremophilanes of the endophytic *Xylaria* sp. BCC 21097 only the α-methylene-lactones **3-42 – 3-44** showed activities in biological assays. Against KB, MCF-7, and NCI-H187 cancer cell lines they exhibited moderate activities in the range of $IC_{50}$ 4-21 μM. Compounds **3-43** and **3-44** were also active against *Plasmodium falciparum* with $IC_{50}$ of 8 and 13 μM respectively and **3-44** was active against *Candida albicans* ($IC_{50}$ 8 μM) (Isaka *et al.*, 2010). From another marine fungus, the tunicate-derived *Penicillium* sp. SS080624SCf1, the secondary metabolites were tested and phomenone **1-2**, sporogen AO-1 **1-3**, and JBIR-28 **3-45** were active against HeLa cells at $IC_{50}$ of 8.3, 19, and 92 μM, respectively (Motohashi *et al.*, 2009). The brominated nor-eremophilane **3-46** from *Penicillium citreonigrum* showed moderate activity of $IC_{50}$ = 16 μg $ml^{-1}$ against the HepG2 cell line (Yuan *et al.*, 2014). The same compound was also found to be active against MDA-MB-231 triple-negative breast cancer (TNBC) cell line ($IC_{50}$ = 5.4 μM) and citreopenin **3-47**, isolated from the same fungus, was modestly active against KB-VIN cell line ($IC_{50}$ = 11 μM) (Yuan *et al.*, 2015).

Progesterone antagonists may have a therapeutic effect in the treatment of breast cancer but RU486, the only available clinical progesterone antagonist, has some side effects. A search for alternative was therefore desirable. The secondary metabolites PF1092A-C **3-48** – **3-50** were discovered in *Penicillium oblatum* and screened for such activity. All three eremophilanes were highly active and inhibited [$^3$H]-progesterone binding to porcine uteri cytosol preparations with $IG_{50}$ of 0.03, 0.22 and 2.2 μM, respectively (Tabata *et al.*, 1997; 1997a). The biological activities of derivatives of PF1092A-C led to the conclusions that only one enantiomer is active and that a large substituent at C-2 may be unfavorable for activity (Kurihara *et al.*, 1997; Tatsuta *et al.*, 1997).

The marine *Cryptosphaeria* sp. CNL-523 produced cryptosphaerolide **4-65,** which was active against a HCT-116 human colon carcinoma cell line with $IC_{50}$ of 4.5 μM (Oh *et al.*, 2010). It was also screened for the inhibition of Mcl-1 (myeloid cell leukemia-1), a protein which belongs to the anti-apoptotic proteins that are a critical negative regulator of apoptosis during myeloblastic leukemia cell differentiation (Fukuchi *et al.*, 2001). Mcl-1 is overexpressed in many cancers, contributes to tumor progression and chemo-resistance and has a critical role in melanoma cell resistance to apoptosis (Boisvert-Adamo *et al.*, 2009). Blocking the activity of Mcl-1 is predicted to initiate apoptosis and sensitize cancer cells to cytotoxic inducers of

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apoptosis (Konopleva *et al.*, 2006). Cryptosphaerolide was found to inhibit the Mcl-1 protein in the Mcl-1/Bak fluorescence resonance energy transfer assay (FRET) screen with an $IC_{50}$ of 11.4 μM. While the methyl ester of this eremophilane had the same activity ($IC_{50}$ = 12.5 μM), the saponified product with the free alcohol at C-1 was neither cytotoxic nor active in the Mcl-1/Bak FRET assay. As for other activities of this type of eremophilanes it underlines the importance of the ester side chain for these activities (Oh *et al.*, 2010) (Figure 5).

The secondary metabolite 07H239-A **4-52** was obtained from a marine-derived xylariaceous fungus LL-07H239 (McDonald *et al.*, 2004) and an endophytic *Xylaria* sp. BL321 (Song *et al.*, 2012a). The compound is mildly cytotoxic (mean $IC_{50}$ = 3.2 μg $ml^{-1}$) with some selectivity for the CCRFCEM human peripheral blood T lymphoblast cell line ($IC_{50}$ = 0.9 μg $ml^{-1}$). In addition, the metabolite 07H239-A was also found to activate α-glucosidase at 0.15 μM (146%). At higher concentrations, however, it inhibited the activity of the enzyme with an $IC_{50}$ of 6.5 μM (Song *et al.*, 2012). In an attempt to improve both the activity and the selectivity of 07H239-A against several cancer cell lines a number of derivatives were generated by amidation of its carboxylic group with several amino acid methyl esters. Most of these derivatives showed enhanced activities compared to the free acid of 07H239-A. Here, glycylmethylate and leucylmethylate derivatives being the most active compounds against the MDA-MB-435 cell line, an activity not shown by 07H239-A (Song *et al.*, 2011).

The saprobic fungus *Berkleasmium nigroapicale* BCC 8220 was the source for berkleasmins A–E and the endophytic fungus *Pestalotiopsis photiniae* of berkleasmin C **4-61** (Yang *et al.*, 2011). Two of these eremophilane esters, berkleasmin A **4-60** and C, exhibited cytotoxic activity against the cancer cell-lines NCI-H187, MCF-7, and KB but also against nonmalignant Vero cells with $IC_{50}$ values of 1.1–7.5 μg $ml^{-1}$. These compounds had also antimalarial activity with $IC_{50}$ of 3.1 and 2.8 μg $ml^{-1}$, respectively (Isaka *et al.*, 2009). Another interesting result of this study was that berkleasmin D and E possessing the same fatty acid side chain as berkleasmin C but being differently oxidized at the eremophilane moiety did not show any activity. This finding points to the importance of the sesquiterpene part for these biological activities. *Chaetoconis* sp. FT087 produced a number of sesquiterpenes, one of them being the trinor-eremophilane dendryphiellin $A_1$ **5-66**. It was found to be moderately active against A2780 and cisplatin resistant A2780CisR cell lines with $IC_{50}$ values of 6.6 and 9.1 μg $ml^{-1}$, respectively (Li *et al.*, 2016).

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The plant-endophytic fungus *Apiognomonia* sp. f24023 produced three eremophilanes with medium-sized C-3 esters. One of them, MBJ-0011 **5-72**, bearing a tetrahydro-α-methyl-5-oxo-2-furanacetic acid moiety, was active against human ovarian adenocarcinoma SKOV-3 cells with $IC_{50}$ of 3.4 μM. On the contrary, MBJ-0012 with the same sesquiterpene but esterified with *trans*,*trans*-3,9-dihydroxy-2-methyl-deca-4,6-dienoic acid showed only weak cytotoxicity ($IC_{50}$ = 63 μM) (Kawahara *et al.*, 2013) (Figure 6).

## Conclusions

Eremophilane sesquiterpenes were first discovered in plants but decades later they have also been reported from fungi. They possess a terpene skeleton, which does not follow the terpene rule of head-to-tail connected isoprene units. Fungal eremophilanes are unique and differ by a sesquiterpene skeleton which is enantiomeric to the one from plants and almost all eremophilanes occurring in fungi are confined to these organisms but are absent in plants. For about half of the described fungal eremophilanes bioactivities have been reported, mainly as phytotoxic, antifungal, antibiotic and anticancer compounds (Table 1).

Most antimicrobials have a rather narrow species range inhibiting either Gram-positive or -negative bacteria but not both. Eremoxylarins are here an exception because the act against both the Gram-positive *Staphylococcus aureus* and the Gram-negative *Pseudomonas aeruginosa* pathogens but it should be kept in mind that eremoxylarin A was also cytotoxic in animal experiments. Because bacteria usually develop resistances against antibiotics rather fast, the search for new anti-infectiva shifts towards blocking pathogenicity factors. In this respect berkleasmin A became interesting because it can restore the sensitivity of aminoglycosides in methicillin-resistant *Staphylococcus aureus* (MRSA). In the field of antivirals integric acid **4-54** should be mentioned, which interacts with the integrase of HIV. Because of the changing lifestyle of humans in the developed countries, obesity becomes an increasing health problem. In a sophisticated assay xylarenals **4-63** – **4-64** have been discovered as specific antagonists of neuropeptide Y and they may become good candidates for the development of antiobesity drugs.

For the treatment of many diseases, a modulation of the immune response of the patient could be desirable. Several periconianones, guignarderemophilanes and acremeremophilanes inhibited NO-production after LPS-stimulation and xylarenones C **4-55**, D **4-56**, F **4-57** and G

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**4-58** were potent inhibitors or reactive oxygen species produced by stimulated neutrophils. Such an activity is often connected with increased cytotoxicity but for several of these compounds only low cytotoxicity was reported.

Fungal eremophilanes do not possess the high activities against cancer cell lines found in other natural products but they may gain importance as drugs for specific aspects in cancer therapies. Regarding the anti-cancer activities of fungal eremophilanes peribysins **3-30** – **3-38** are remarkable. These compounds are cell-adhesion inhibitors and act probably via inhibition of tyrosine kinases. Inhibition of cell adhesion is an interesting characteristic for the prevention of the spreading of metastases. For some breast cancers the control of progesterone signaling is important. Here, PF1092A-C **3-48** – **3-50** from *Penicillium oblatum* may become an interesting alternative for the existing progesterone antagonists.

Several fungal eremophilanes carry unusual carboxylic acids at C-1 or C-3. Many of these fatty acids are branched, some of them bear in addition hydroxyl groups. Only in one study, the biosynthesis of these fatty acids has been elucidated and it has been demonstrated that the methylation of the fatty acid backbone was achieved by using methionin similar to the biosynthesis of tenellin (Eley *et al.*, 2007). An alternative pathway proceeding through an iterative type I polyketide synthase produces the required poly-unsaturated methylated heptaketide for fusarin C is also possible (Song *et al.*, 2004). Looking at the different structures of the carboxylic acids at C-1 in Figure 5, their limited structure diversity become apparent. This may be an indication that these esters have an important role in the bioactivities of these eremophilanes. Structure-activity studies have confirmed this for some compounds but for others the structure of the ester side chain was less important. While a slight variation in the ester moiety leads to half of the activity of eremoxylarin B **4-53** compared to eremoxylarin A **4-51,** the carboxylic acid of integric acid **4-54** alone has considerable activity and the saponification of the C-1 ester of cryptosphaerolide **4-65** abolished all activity, no effect of the ester side chain of the berkleasmins could be found.

The observation that a given compounds shows activity in a given test is only the first indication that such a compound could be brought into application. The cytotoxic activity of fungal eremophilanes is mostly moderate and several metabolites from other substance classes are more active or more selective. These eremophilanes are therefore facing a strong competition to be selected for further improvement. Novel isolates from novel and

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underexplored habitats will certainly lead to new eremophilanes and their characteristics will hopefully shed also more light on the role of the unusual carboxylic acids at C-1 or C-3.

## Outlook

Secondary metabolites possess often rather sophisticated mode of actions but they are not optimized for agricultural or medical applications. Chemical modifications are usually the key for optimizing the compound for the special needs of drug delivery, improved production and the suppression of undesirable side effects. Diversity Oriented Synthesis (DOS) uses natural compounds as lead compounds and generate substance libraries from these compounds for further screening and testing (Burke and Schreiber, 2004). The identification of the target molecule allows the establishment of quantitative structure-activity relationships (QSAR) which will facilitate the design of highly active and specific drugs. Standardization of methods and protocols is here important for the comparison of results (Gandhi and Morris, 2009).

An interesting alternative to the generation of new isolates and the screening of their natural products is a synthetic biology approach (Zebec *et al.*, 2016). As more and more sesquiterpene cyclases are characterized our insight in their reaction mechanisms improves (Christianson, 2006). The stabilization of the flexible substrate farnesylpyrophosphate in the reaction centers of the enzymes is responsible for the specificity of the enzymes. Water bound at the active site of the enzyme seems to control the folding of the substrate and the chemistry of the product (Chen *et al.*, 2016) but terpene cyclases have been demonstrated to be not very substrate specific (Vattekkatte *et al.*, 2017). Aristolochene synthase from *Aspergillus terreus*, however, shows high specificity and X-ray crystal structure analysis of the enzyme revealed the evolution of the active site towards the formation of a single sesquiterpene skeleton (Shishova *et al.*, 2007). This, however, opens the door for targeted engineering of the enzymes in order to get novel products or to increase the production of specific sesquiterpenes (Back and Chappell, 1996). Fungal eremophilane synthases and mutants thereof have been isolated and characterized (Burkhardt *et al.*, 2016). Furthermore, the heterologous expression of terpenes synthases (Leavell *et al.*, 2016; Wu *et al.*, 2016), the combination of enzymes from different microorganisms and different pathways (Khosla *et al.*, 2003) or the genetic engineering of sesquiterpene cyclases can be used for the generation of novel compounds (Ajikumar *et al.*, 2016). Using unnatural substrates can broaden the product spectrum but

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even for well characterized cyclases the outcome is difficult to predict (Hammer *et al.*, 2016). Heterologous expression of enzymes has its limitation in the compatibility with other enzyme reactions or the toxicity of novel compounds to the producing cells. This limitation can be overcome by *in vitro* systems, e.g. using enzymes in continuous-flow biocatalysis or on microarrays (Britton *et al.*, 2017). The resulting substance libraries can then be screened for their bioactivities (Kwon *et al.*, 2012).

Finding novel bioactive compounds is only the first step on a long and expensive way to clinical applications. Before any clinical trials ADMET (absorption, distribution, metabolism, and excretion - toxicity in pharmacokinetics) of the potential drug is needed. To reduce the costs miniaturized 3D cell-culture arrays for high-throughput toxicity screening of drug candidates can be applied. Here, compounds are screened against human cells attached to a functionalized glass slide in combination with another array carrying a mixture of three P450-oxidases designed to emulate the human liver. Such a device gives both the activity of the compounds and the toxicity of their metabolites (Lee *et al.*, 2008). However, all these *in vitro* results have to be validated in animal models and finally in clinical studies. Improvements in the generation and testing of novel drug candidates can help to reduce the immense costs the successful development of a new drug consumes. It has been estimated that the average costs for the development of a new pharmaceutical drug were 2-3 billion US-$ in 2016 (DiMasi *et al.*, 2016; Nosengo, 2016). Relying on natural compounds already optimized in the evolutionary process can help to shape the immune system or to combat specific forms of cancer, only to mention some of the most urgent problems.

## Acknowledgments

K.T.Y. acknowledges the support by a stipend of the “Science without border (Ciências sem Fronteiras)” program of Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) from Brazil and the German Academic Exchange Service (DAAD). D.F. is thankful for a stipend of CAPES.

## Conflict of interest statement

The authors declare that there is no conflict of interest.

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## References


Ajikumar, P.K., Stephanopoulos, G., and Too, H.P. (2016). Microbial engineering for the production of chemical and pharmaceutical products from the isoprenoid pathway. U.S. Patent No. 9,359,624.

Angeles, A.R., Dorn, D.C., Kou, C.A., Moore, M.A., and Danishefsky, S.J. (2007). Total synthesis of peribysin E necessitates revision of the assignment of its absolute configuration. Angew. Chem. Int. Ed. *46*, 1451-1454.

Back, K., and Chappell, J. (1996). Identifying functional domains within terpene cyclases using a domain-swapping strategy. Proc. Natl. Acad. Sci. USA *93*, 6841-6845.

Bohlmann, F., Knoll, K.H., Zdero, C., Mahanta, P.K., Grenz, M., Suwita, A., Ehlers, D., LeVan, N., Abraham, W.-R., and Natu, A.A. (1977). Terpen-Derivate aus *Senecio* Arten. Phytochemistry *16*, 965-985.

Boisvert-Adamo, K., Longmate, W., Abel, E.V., and Aplin, A.E. (2009). Mcl-1 is required for melanoma cell resistance to anoikis. Mol. Cancer Res. *7*, 549-556.

Bradfield, A.E., Penfold, A.R., and Simonsen, J.L. (1932). The constitution of eremophilone and of two related hydroxy-ketones from the wood oil of *Eremophila mitchelli*. J. Chem. Soc. (Resumed), 2744-2759.

Britton, J., Dyer, R.P., Majumdar, S., Raston, C.L., and Weiss, G.A. (2017). Ten‐minute protein purification and surface tethering for continuous‐flow biocatalysis. Angew. Chem. Int. Ed. *56*, 2296-2301.

Burke, M.D., and Schreiber, S.L. (2004). A planning strategy for diversity-oriented synthesis. Angew. Chem. Int. Ed. *43*, 46-58.

Burkhardt, I., Siemon, T., Henrot, M., Studt, L., Rösler, S., Tudzynski, B., Christmann, M., and Dickschat, J.S. (2016). Mechanistic characterisation of two sesquiterpene cyclases from the plant pathogenic fungus *Fusarium fujikuroi*. Angew. Chem. Int. Ed. *55*, 8748-8751.

Chang, J.C., Hsiao, G., Lin, R.K., Kuo, Y.H., Ju, Y.M., and Lee, T.H. (2017). Bioactive constituents from the termite nest-derived medicinal fungus *Xylaria nigripes*. J. Nat. Prod. *80*, 38-44.

Che, Y., Gloer, J.B., and Wicklow, D.T. (2002). Phomadecalins A–D and phomapentenone A: new bioactive metabolites from *Phoma* sp. NRRL 25697, a fungal colonist of *Hypoxylon* stromata. J. Nat. Prod. *65*, 399-402.

Chen, M., Chou, W.K., Al-Lami, N., Faraldos, J.A., Allemann, R.K., Cane, D.E., and Christianson, D.W. (2016). Probing the role of active site water in the sesquiterpene cyclization reaction catalyzed by aristolochene synthase. Biochemistry *55*, 2864-2874.

Cheng, Z., Zhao, J., Liu, D., Proksch, P., Zhao, Z., and Lin, W. (2016). Eremophilane-type sesquiterpenoids from an *Acremonium* sp. fungus isolated from deep-sea sediments. J. Nat. Prod. *79*, 1035-1047.

Christianson, D.W. (2006). Structural biology and chemistry of the terpenoid cyclases. Chem. Rev. *106*, 3412-3442.

Daengrot, C., Rukachaisirikul, V., Tansakul, C., Thongpanchang, T., Phongpaichit, S., Bowornwiriyapan, K., and Sakayaroj, J. (2015). Eremophilane sesquiterpenes and


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diphenyl thioethers from the soil fungus *Penicillium copticola* PSU-RSPG138. J. Nat. Prod. *78*, 615-622.

DiMasi, J.A., Grabowski, H.G., and Hansen, R.W. (2016). Innovation in the pharmaceutical industry: new estimates of R&D costs. J. Health Econ. *47*, 20-33.

Ebel, R. (2010). Terpenes from marine-derived fungi. Mar. Drugs *8*, 2340-2368.

Efremenkova, O.V., Vasiljeva, B.F., Zenkova, V.A., Korolev, A.M., Lusikov, Y.N., Efimenko, T.A., Malanicheva, I.A., Mirchink, E.P., Isakova, E.B., Bilanenko, E.N., and Kamzolkina, O.V. (2015). Antimicrobial properties of eremoxylarin A produced by ascomycete of Sordariomycetes in submerged culture. Antibiot. Khimioter. *60*, 23-28.

Eley, K.L., Halo, L.M., Song, Z., Powles, H., Cox, R.J., Bailey, A.M., Lazarus, C.M., and Simpson, T.J. (2007). Biosynthesis of the 2‐pyridone tenellin in the insect pathogenic fungus *Beauveria bassiana*. ChemBioChem *8*, 289-297.

Ferretti, J.J., Gilmore, K.S., and Courvalin, P. (1986). Nucleotide sequence analysis of the gene specifying the bifunctional 6'-aminoglycoside acetyltransferase 2"-aminoglycoside phosphotransferase enzyme in *Streptococcus faecalis* and identification and cloning of gene regions specifying the two activities. J. Bacteriol. *167*, 631-638.

Fukuchi, Y., Kizaki, M., Yamato, K., Kawamura, C., Umezawa, A., Hata, J., Nishihara, T., and Ikeda, Y. (2001). Mcl-1, an early-induction molecule, modulates activin A-induced apoptosis and differentiation of CML cells. Oncogene *20*, 704-713.

Gandhi, Y.A., and Morris, M.E. (2009). Structure–activity relationships and quantitative structure–activity relationships for breast cancer resistance protein (ABCG2). AAPS J. *11*, 541.

García-Estrada, C., and Martín, J.F. (2016). Biosynthetic gene clusters for relevant secondary metabolites produced by *Penicillium roqueforti* in blue cheeses. Appl. Microbiol. Biotechnol. *100*, 8303-8313.

Gorst-Allman, C.P., and Steyn, P.S. (1982). Biosynthesis of PR toxin by *Penicillium roqueforti*, Part 2 evidence for a hydride shift from 2H NMR spectroscopy. Tetrahedron Lett. *23*, 5359-5362.

Gubiani, J.R., Zeraik, M.L., Oliveira, C.M., Ximenes, V.F., Nogueira, C.R., Fonseca, L.M., Silva, D.H.S., Bolzani, V.S., and Araujo, A.R. (2014). Biologically active eremophilane-type sesquiterpenes from *Camarops* sp., an endophytic fungus isolated from *Alibertia macrophylla*. J. Nat. Prod. *77*, 668-672.

Gubiani, J.R., Nogueira, C.R., Pereira, M.D.P., Young, M.C.M., Ferreira, P.M.P., Moraes, M.O., Pessoa, C., Bolzani, V.S., and Araujo, A.R. (2016). Rearranged sesquiterpenes and branched polyketides produced by the endophyte *Camarops* sp. Phytochem. Lett. *17*, 251–257.

Guerriero, A., D'Ambrosio, M., Pietra, F., Cuomo, V. and Vanzanella, F. (1988), Dendryphiellin A, the first fungal trinor-eremophilane. Isolation from the marine deuteromycete *Dendryphiella salina* (Sutherland) Pugh et Nicot. Helv. Chim. Acta *71*, 57–61.

Hammer, S.C., Syrén, P.O., and Hauer, B. (2016). Substrate pre‐folding and water molecule organization matters for terpene cyclase catalyzed conversion of unnatural substrates. Chemistry Select *1*, 3589-3593.

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Download Date | 9/5/17 2:55 AM

Hatakeyama, T., Koseki, T., Murayama, T., and Shiono, Y. (2010). Eremophilane sesquiterpenes from the endophyte *Microdiplodia* sp. KS 75-1 and revision of the stereochemistries of phomadecalins C and D. Phytochem. Lett. *3*, 148-151.

Höller, U., König, G.M., and Wright, A.D. (1999). Three new metabolites from marine-derived fungi of the genera *Coniothyrium* and *Microsphaeropsis*. J. Nat. Prod. *62*, 114-118.

Hou, C., Kulka, M., Zhang, J., Li, Y., and Guo, F. (2014). Occurrence and biological activities of eremophilane-type sesquiterpenes. Mini Rev. Med. Chem. *14*, 664-677.

Hu, Z.Y., Li, Y.Y., Huang, Y.J., Su, W.J., and Shen, Y.M. (2008). Three new sesquiterpenoids from *Xylaria* sp. NCY2. Helv. Chim. Acta *91*, 46-52.

Hu, Z.Y., Li, Y.Y., Lu, C.H., Lin, T., Hu, P., and Shen, Y.M. (2010). Seven novel linear polyketides from *Xylaria* sp. NCY2. Helv. Chim. Acta *93*, 925-933.

Isaka, M., Srisanoh, U., Veeranondha, S., Choowong, W., and Lumyong, S. (2009). Cytotoxic eremophilane sesquiterpenoids from the saprobic fungus *Berkleasmium nigroapicale* BCC 8220. Tetrahedron *65*, 8808-8815.

Isaka, M., Chinthanom, P., Boonruangprapa, T., Rungjindamai, N., and Pinruan, U. (2010). Eremophilane-type sesquiterpenes from the fungus *Xylaria* sp. BCC 21097. J. Nat. Prod. *73*, 683-687.

Jayasuriya, H., Zink, D.L., Polishook, J.D., Bills, G.F., Dombrowski, A.W., Genilloud, O., Pelaez, F.F., Herranz, L., Quamina, D., Lingham, R.B., Danzeizen, R., Graham, P.L., Tomassini, J.E., and Singh, S.B. (2005). Identification of diverse microbial metabolites as potent inhibitors of HIV‐1 Tat transactivation. Chem. Biodiv. *2*, 112-122.

Kawahara, T., Itoh, M., Izumikawa, M., Sakata, N., Tsuchida, T., and Shin-ya, K. (2013). Three eremophilane derivatives, MBJ-0011, MBJ-0012 and MBJ-0013, from an endophytic fungus *Apiognomonia* sp. f24023. J. Antibiot. *66*, 299-302.

Kawamura, H., Pulici, M., Koshino, H., Esumi Y., Uzawa, J., Kumagai, H., and Sugawara, F. (2000). New potent immunosuppressive eremophilane isolated from *Trichothecium roseum*. Nat. Prod. Lett. *14*, 299-304.

Khosla, C., and Keasling, J.D. (2003). Metabolic engineering for drug discovery and development. Nat. Rev. Drug Discov. *2*, 1019-1025.

Konopleva, M., Contractor, R., Tsao, T., Samudio, I., Ruvolo, P.P., Kitada, S., Deng, X., Zhai, D., Shi, Y.X., Sneed, T., Verhaegen, M., Soengas, M., Ruvolo, V.R., McQueen, T., Schober, W.D., Watt, J.C., Jiffar, T., Ling, X., Marini, F.C., Harris, D., Dietrich, M., Estrov, Z., McCubrey, J., May, W.S., Reed, J.C., and Andreeff, M. (2006). Mechanisms of apoptosis sensitivity and resistance to the BH3 mimetic ABT-737 in acute myeloid leukemia. Cancer Cell *10*, 375-388.

Koshino, H., Satoh, H., Yamada, T., and Esumi, Y. (2006). Structural revision of peribysins C and D. Tetrahedron Lett. *47*, 4623-4626.

Kurihara, K. I., Tanabe, K., Shinei, R., Okonogi, T., Ohtsuka, Y., Omoto, S., Yasuda, S., and Tatsuta, K. (1997). Synthesis of (+-)-PF1092A, B, and C; new nonsteroidal progesterone receptor ligands. J. Antibiot. *50*, 360-362.

Kwon, S.J., Mora-Pale, M., Lee, M.Y., and Dordick, J.S. (2012). Expanding nature's small molecule diversity via *in vitro* biosynthetic pathway engineering. Curr. Opin. Chem. Biol. *16*, 186-195.

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Leavell, M.D., McPhee, D.J., and Paddon, C.J. (2016). Developing fermentative terpenoid production for commercial usage. Curr. Opin. Biotechnol. *37*, 114-119.

Lee, M.Y., Kumar, R.A., Sukumaran, S.M., Hogg, M.G., Clark, D.S., and Dordick, J.S. (2008). Three-dimensional cellular microarray for high-throughput toxicology assays. Proc. Natl. Acad. Sci. USA *105*, 59-63.

Li, C.-S., Ding, Y., Yang, B.-J., Hoffman, N., Yin, H.-Q., Mahmud, T., Turkson, J., Cao, S. (2016). Eremophilane sesquiterpenes from Hawaiian endophytic fungus *Chaetoconis* sp. FT087. Phytochemistry *126*, 41-46.

Lin, A., Wu, G., Gu, Q., Zhu, T., and Li, D. (2014). New eremophilane-type sesquiterpenes from an Antarctic deep-sea derived fungus, *Penicillium* sp. PR19 N-1. Arch. Pharm. Res. *37*, 839-844.

Liu, Y., Li, Y., Qu, J., Ma, S., Zang, C., Zhang, Y., and Yu, S. (2015). Eremophilane sesquiterpenes and polyketones produced by an endophytic *Guignardia* fungus from the toxic plant *Gelsemium elegans*. J. Nat. Prod. *78*, 2149-2154.

Liu, J., Zhang, D., Zhang, M., Zhao, J., Chen, R., Wang, N., Zhang, D., and Dai, J. (2016). Eremophilane sesquiterpenes from an endophytic fungus *Periconia* species. J. Nat. Prod. *79*, 2229-2235.

McDonald, L.A., Barbieri, L.R., Bernan, V.S., Janso, J., Lassota, P., and Carter, G.T. (2004). 07H239-A, a new cytotoxic eremophilane sesquiterpene from the marine-derived Xylariaceous fungus LL-07H239. J. Nat. Prod. *67*, 1565-1567.

Mierau, V., Anke, T., and Sterner, O. (2003). Dacrymenone and VM 3298-2--new antibiotics with antibacterial and antifungal activity. Z. Naturforsch C *58*, 541-546.

Miller, D.J., Gao, J., Truhlar, D.G., Young, N.J., Gonzalez, V., and Allemann, R.K. (2008). Stereochemistry of eudesmane cation formation during catalysis by aristolochene synthase from *Penicillium roqueforti*. Org. Biomol. Chem. *6*, 2346-2354.

Moulé, Y., Jemmali, M., and Rousseau, N. (1976). Mechanism of the inhibition of transcription by PR toxin, a mycotoxin from *Penicillium roqueforti*. Chem. Biol. Interact. *14*, 207-216.

Moreau, S., Lablache-Combier, A., and Biguet, J. (1980). Production of eremofortins A, B, and C relative to formation of PR toxin by *Penicillium roqueforti*. Appl. Environ. Microbiol. *39*, 770-776.

Moreau, S., Lablache-Combier, A., and Biguet, J. (1981). Biosynthesis of PR toxin from [1, 2-$^{13}$C] acetate: occurrence of induced $^{13}$C-$^{13}$C coupling. Phytochemistry *20*, 2339-2342.

Motohashi, K., Hashimoto, J., Inaba, S., Khan, S. T., Komaki, H., Nagai, A., Takagi, M., and Shin-ya, K. (2009). New sesquiterpenes, JBIR-27 and-28, isolated from a tunicate-derived fungus, *Penicillium* sp. SS080624SCf1. J. Antibiot. *62*, 247-250.

Neuenschwander, M., Neuenschwander, A., Steinegger, E., and Engel, P. (1979). Struktur der Sesquiterpene von *Petasites hybridus* (L.) GM et SCH.: Petasol‐ und Isopetasol‐Abkömmlinge. Helv. Chim. Acta *62*, 609-626.

Nicoletti, R., and Trincone, A. (2016). Bioactive compounds produced by strains of *Penicillium* and *Talaromyces* of marine origin. Mar. Drugs *14*, 37.

Nosengo, N. (2016). New tricks for old drugs. Nature *534*, 314-316.

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Download Date | 9/5/17 2:55 AM

Ogasawara, Y., Yoshida, J., Shiono, Y., Miyakawa, T., and Kimura, K-I. (2008). New eremophilane sesquiterpenoid compounds, eremoxylarins A and B directly inhibit Calcineurin in a manner independent of Immunophilin. J. Antibiot. *61*, 496–502.

Oh, H., Jensen, P.R., Murphy, B.T., Fiorilla, C., Sullivan, J.F., Ramsey, T., and Fenical, W. (2010). Cryptosphaerolide, a cytotoxic Mcl-1 inhibitor from a marine-derived ascomycete related to the genus *Cryptosphaeria*. J. Nat. Prod. *73*, 998-1001.

Okabe, M., Kawamura, K., Miyagishima, T., Itaya, T., Goodwyn, D., Shoji, M., Vogler, R.W., Sakurada, K., Uehara, M., and Miyazaki, T. (1994). Effect of herbimycin A, an inhibitor of tyrosine kinase, on protein tyrosine kinase activity and phosphotyrosyl proteins of Ph1-positive leukemia cells. Leukemia Res. *18*, 213-220.

Oliveira, C.M.D., Silva, G.H., Regasini, L.O., Flausino Jr, O., López, S.N., Abissi, B.M., Gomes de Zouza-Berlinck, B.M.G., Sette, L.D., Bonugli-Santos, R.C., Rodrigues, A., da Silva-Bolzani, V., and Araujo, A.R. (2011). Xylarenones C-E from an endophytic fungus isolated from *Alibertia macrophylla*. J. Nat. Prod. *74*, 1353-1357.

Penfold, A.R., and Simonsen, J.L. (1939). The constitutions of eremophilone, hydroxyeremophilone, and hydroxydihydroeremophilone. Part III. J Chem. Soc. (Resumed), 87-89.

Pérez-Castorena, A.L., Arciniegas, A., Guzmán, S.L., Villaseñor, J.L., and Romo de Vivar, A. (2006). Eremophilanes from *Senecio mairetianus* and some reaction products. J. Nat. Prod. *69*, 1471-1475.

Pinder, A.R. (1977). The chemistry of the eremophilane and related sesquiterpenes. Prog. Chem. Org. Nat. Prod. *34*, 81-186.

Riche, C., Pascard-Billy, C., Devys, M., Gaudemer, A., Barbier, M. and Bousquet, J.F. (1974) Structure cristalline et moleculaire de la phomenone, phytotoxine produite par le champignon *Phoma exigua var. non oxydabilis*. Tetrahedron Lett. *32*, 2765-2766.

Riclea, R., and Dickschat, J.S. (2015). Identification of intermediates in the biosynthesis of PR toxin by *Penicillium roqueforti*. Angew. Chem. Int. Ed. *54*, 12167-12170.

Schifrin, A., Ly, T.T., Günnewich, N., Zapp, J., Thiel, V., Schulz, S., Hannemann, F., Khatri, Y., and Bernhardt, R. (2015). Characterization of the gene cluster CYP264B1‐geoA from *Sorangium cellulosum* So ce56: biosynthesis of (+)‐eremophilene and its hydroxylation. ChemBioChem *16*, 337-344.

Schneider, G., Anke, H., and Sterner, O. (1997). New Secondary metabolites from a mycophilic *Hansfordia* species. Nat. Prod. Lett. 10, 133-138.

Shiono, Y., and Murayama, T. (2005). New eremophilane-type sesquiterpenoids, eremoxylarins A and B from xylariaceous endophytic fungus YUA-026. Z. Naturforsch. B *60*, 885-890.

Shishova, E.Y., Di Costanzo, L., Cane, D.E., and Christianson, D.W. (2007). X-ray crystal structure of aristolochene synthase from *Aspergillus terreus* and the evolution of templates for the cyclization of farnesyl diphosphate. Biochemistry *46*, 1941-1951.

Silva, G.H., Oliveira, C.M.D., Teles, H.L., Pauletti, P.M., Castro-Gamboa, I., Silva, D.H.S., Bolzani, V.S., Young, M.C.M., Costa-Neto, C.M., Pfenning, L.H., Berlinck, R.G.S., and Araujo, A.R. (2010). Sesquiterpenes from *Xylaria* sp., an endophytic fungus associated with *Piper aduncum* (Piperaceae). Phytochem. Lett. *3*, 164-167.

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Singh, S.B., Zink, D., Polishook, J., Valentino, D., Shafiee, A., Silverman, K., Felock, P., Teran, A., Vilella, D., Hazuda, D.J., and Lingham, R.B. (1999). Structure and absolute stereochemistry of HIV-1 integrase inhibitor integric acid. A novel eremophilane sesquiterpenoid produced by a *Xylaria* sp. Tetrahedron Lett. *40*, 8775-8779.

Singh, S.B., Felock, P., and Hazuda, D.J. (2000). Chemical and enzymatic modifications of integric acid and HIV-1 integrase inhibitory activity. Bioorg. Med. Chem. Lett. *10*, 235-238.

Smith, C.J., Morin, N.R., Bills, G.F., Dombrowski, A.W., Salituro, G.M., Smith, S.K., Zhao, A., and MacNeil, D.J. (2002). Novel sesquiterpenoids from the fermentation of *Xylaria persicaria* are selective ligands for the NPY Y5 receptor. J. Org. Chem. *67*, 5001-5004.

Song, Z., Cox, R. J., Lazarus, C. M., and Simpson, T. J. (2004). Fusarin C biosynthesis in *Fusarium moniliforme* and *Fusarium venenatum*. ChemBioChem *5*, 1196-1203.

Song, Y.X., Cheng, B., Zhu, X., Qiao, L.T., Wang, J.J., Gu, Y.C., Li, M.F., Liu, L., and Lin, Y.C. (2011). Synthesis and cytotoxic evaluation of eremophilane sesquiterpene 07H239-A derivatives. Chem. Pharm. Bull. *59*, 1186-1189.

Song, Y., Wang, J., Huang, H., Ma, L., Wang, J., Gu, Y., Liu, L., and Lin, Y. (2012). Four eremophilane sesquiterpenes from the mangrove endophytic fungus *Xylaria* sp. BL321. Marine Drugs *10*, 340-348.

Song, Y.X., Wang, J., Li, S.W., Cheng, B., Li, L., Chen, B., Liu, L., Lin, Y.C., and Gu, Y.C. (2012a) Metabolites of the mangrove fungus *Xylaria* sp. BL321 from the South China Sea. Planta Medica *78*, 172-176.

Srisapoomi, T., Ichiyanagi, T., Nakajima, H., Aimi, T., and Boonlue, S. (2015). Biosynthesis of integric acid isolated from the wood-decay fungus *Xylaria feejeensis* 2FB-PPM08M. Curr. Microbiol. *70*, 550-555.

Srisapoomi, T. Ichiyanagi, T., Nakajima, H., Aimi, T., and Boonlue, S. (2015a). Biological activities of integric acid isolated from the wood-decay fungus *Xylaria feejeensis* 2FB-PPM08M. Chiang Mai J. Sci. *42*, 70-78.

Suga, T., Shiina, M., Asami, Y., Iwatsuki, M., Yamamoto, T., Nonaka, K., Masuma, R., Matsiu, H., Hanaki, H., Iwamoto, S., Onodera, H., Shiomi, K., and Omura, S. (2016). Paraphaeosphaeride D and berkleasmin F, new circumventors of arbekacin resistance in MRSA, produced by *Paraphaeosphaeria* sp. TR-022. J. Antibiot. *69*, 605-610.

Tabata, Y., Hatsu, M., Kurata, Y., Miyajima, K., Tani, M., Sasaki, T., Kodama, Y.,Tsuruoka, T., and Omoto, S. (1997). PF1092A, B and C, new nonsteroidal progesterone receptor ligands produced by *Penicillium oblatum*. J. Antibiot. *50*, 309-313.

Tabata, Y., Miike, M., Hatsu, M., Kurata, Y., Yaguchi, T., Someya, A., Miyadoh, S., Hoshico, S., Tsuruoka, T., and Omoto, S. (1997a). PF1092A, B and C, new nonsteroidal progesterone receptor ligands produced by *Penicillium oblatum*. I. Taxonomy of producing strain, fermentation, isolation and biological activities. J. Antibiot. *50*, 304-308.

Tanaka, S., Wada, K., Marumo, S., and Hattori, H. (1984). Structure of sporogen-AO 1, a sporogenic substance of *Aspergillus oryzae*. Tetrahedron Lett. *25*, 5907-5910.

Tatsuta, K., Yasuda, S., Kurihara, K.I., Tanabe, K., Shinei, R., and Okonogi, T. (1997). Total synthesis of progesterone receptor ligands,(-)-PF1092A, B and C. Tetrahedron Lett. *38*, 1439-1442.

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Vattekkatte, A., Garms, S., and Boland, W. (2017). Alternate cyclization cascade initiated by substrate isomer in multiproduct terpene synthase from *Medicago truncatula*. J. Org. Chem. *82*, 2855-2861.

Wallach, O. (1887). Zur Kenntnis der Terpene und der ätherischen Öle. Liebig‘s Ann. Chem. *239*, 1-54.

Wang, Y.T., Xue, Y.R., and Liu, C.H. (2015). A brief review of bioactive metabolites derived from deep-sea fungi. Mar. Drugs *13*, 4594-4616.

Wang, L., Li, M., Tang, J., and Li, X. (2016). Eremophilane sesquiterpenes from a deep marine-derived fungus, *Aspergillus* sp. SCSIOW2, cultivated in the presence of epigenetic modifying agents. Molecules *21*, 473.

Weber, C., Negrescu, E., Erl, W., Piersch, A., Frankenberger, M., Ziegler-Heitbrock, H.W.L., Siess, W., and Weber, P.C. (1995). Inhibitors of protein tyrosine kinase suppress TNF-stimulated induction of endothelial cell adhesion molecules. J. Immunol. *155*, 445-451.

Wu, G., Lin, A., Gu, Q., Zhu, T., and Li, D. (2013). Four new chloro-eremophilane sesquiterpenes from an Antarctic deep-sea derived fungus, *Penicillium* sp. PR19N-1. Marine Drugs *11*, 1399-1408.

Wu, W., Tran, W., Taatjes, C. A., Alonso-Gutierrez, J., Lee, T. S., and Gladden, J. M. (2016). Rapid discovery and functional characterization of terpene synthases from four endophytic xylariaceae. PLoS One *11*, e0146983.

Yamada, T., Iritani, M., Minoura, K., Kawai, K., and Numata, A. (2004). Peribysins A–D, potent cell-adhesion inhibitors from a sea hare-derived culture of *Periconia* species. Org. Biomol. Chem. *2*, 2131-2135.

Yamada, T., Doi, M., Miura, A., Harada, W., Hiramura, M., Minoura, K., Tanaka, R., and Numata, A. (2005). Absolute stereostructures of cell-adhesion inhibitors, peribysins A, E, F and G, produced by a sea hare-derived *Periconia* sp. J. Antibiotics *58*, 185-191.

Yamada, T., Minoura, K., Tanaka, R., and Numata, A. (2006). Cell-adhesion inhibitors produced by a sea hare-derived *Periconia* sp. II Absolute stereostructures of peribysins H and I. J. Antibiot. *59*, 345–350.

Yang, X.L., Zhang, S., Zhu, H.J., and Luo, D.Q. (2011). Dihydroberkleasmin A: a new eremophilane sesquiterpenoid from the fermentation broth of the plant endophytic fungus *Pestalotiopsis photiniae*. Molecules *16*, 1910-1916.

Yuan, W.H., Wei, Z.W., Dai, P., Wu, H., Zhao, Y.X., Zhang, M.M., Jiang, N., and Zheng, W.F. (2014). Halogenated metabolites isolated from *Penicillium citreonigrum*. Chem. Biodivers. *11*, 1078-1087.

Yuan, W.H., Goto, M., Hsieh, K.Y., Yuan, B., Zhao, Y., Morris-Natschke, S.L., and Lee, K.H. (2015). Selective cytotoxic eremophilane-type sesquiterpenes from *Penicillium citreonigrum*. J. Asian Nat. Prod. Res. *17*, 1239-1244.

Zebec, Z., Wilkes, J., Jervis, A.J., Scrutton, N.S., Takano, E., and Breitling, R. (2016). Towards synthesis of monoterpenes and derivatives using synthetic biology. Curr. Opin. Chem. Biol. *34*, 37-43.

Zhang, D., Ge, H., Zou, J. H., Tao, X., Chen, R., and Dai, J. (2014). Periconianone A, a new 6/6/6 carbocyclic sesquiterpenoid from endophytic fungus *Periconia* sp. with neural anti-inflammatory activity. Org. Lett. *16*, 1410-1413.

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## Tables and figures

**Table 1** List of the producing fungi and medicinal activity of their eremophilanes.

| Producing fungus | Active metabolite | Activity | Reference |
|---|---|---|---|
| *Acremonium* sp. Isolated from deep-sea sediments | Acremeremophilanes B **5-67**, C **5-68**, D **5-69**, E **5-70**, F **5-71** | Inhibitory activity against the lipopolysaccharide (LPS)-induced nitric oxide (NO) | Cheng *et al.*, 2016 |
| *Apiognomonia* sp. F24023 | MBJ-0011 **5-72** | Antitumor activity | Kawahara *et al.*, 2013 |
| *Aspergillus oryzae*, marine *Penicillium* sp. SS080624SCfl, *Penicillium copticola* PSURSPG138, *Penicillium* sp. | Sporogen AO I **1-3** | Antimalarial activity, inhibitor of the regulatory protein HIV-1 and cytotoxic activity | Schneider *et al.*, 1997; Tanaka *et al.*, 1984; Daengrot *et al.*, 2015; Jayasuriya *et al.*, 2005; Motohashi *et al.*, 2009 |
| *Aspergillus* sp. SCSIOW2 | Dihydrobipolaroxin **2-17**, Dihydrobipolaroxin B **2-18**, C **2-19**, D **2-20** | Weak nitric oxide inhibitory activity without cytotoxic effects | Wang *et al.*, 2016 |
| *Berkleasmium nigroapicale* BCC 8220 *Pestalotiopsis photiniae* | Berkleasmin A **4-60**, C **4-61** | Cytotoxic, antitumor and antimalarial activity | Yang *et al.*, 2011 Isaka *et al.*, 2009 |
| *Camarops* sp. | Xylarenone C **4-55**, D **4-56**, F **4-57**, G **4-58** | Inhibitors of ROS (**4-55**, **4-56**, **4-57**, **4-58**) and cytotoxic activity (**4-55**, **4-56**) | Gubiani *et al.*, 2014 Gubiani *et al.*, 2016 |
| *Chaetoconis* sp. FT087 *Dendryphiella salina* | Trinor-eremophilane dendryphiellin $A_1$ **5-66** | Cytotoxic activity | Li *et al.*, 2016 Guerriero *et al.*, 1988 |
| *Cryptosphaeria* sp. CNL-523 | Cryptosphaerolide **4-65** | Cytotoxic activity | Fukuchi *et al.*, 2001; Boisvert-Adamo *et al.*, 2009; Konopleva *et al.*, 2006; Oh *et al.*, 2010 |
| *Dacrymyces* sp. | Dacrymenone **1-13** | Antifungal and antibacterial activity | Mierau *et al.*, 2003 |
| *Drechslera gigantea*, endophytic *Xylaria* sp., *Penicillium* sp. SS080624SCfl, *Phoma exigua var non oxydabilis* | Phomenone **1-2** | Cytotoxic activity Antifungal activity | Riche *et al.*, 1974; Motohashi *et al.*, 2009; Silva *et al.*, 2010; |
| Endophytic fungus | Xylarenone C **4-55**, Xylarenone A **1-6** | Protease inhibitory activity | Oliveira *et al.*, 2011 |
| *Guignardia mangiferae* | Guignarderemophilane A **2-27**, C **2-28**, D **2-29** | Anti-inflammatory activity | Liu *et al.*, 2015 |
| *Microdiplodia* sp. KS 75-1 | Phomadecalins D **1-10** Dihydro-xylarenone A **1-11**, acetate **1-12** | Antibacterial activity | Che *et al.*, 2002 |
| *Microsphaeropsis* sp. | Microsphaeropsisin **1-14** | Weak antibacterial activity and antifungal activity | Höller *et al.*, 1999 |
| *Paraphaeosphaeria* sp. TR-022 | Berkleasmin A **4-60**, F **4-62** | Resensitizing MRSA against aminoglycosides (**4-60**, **4-62**) Cytotoxic, antitumor and antimalarial activity (**4-60**) | Suga *et al.*, 2016 Isaka *et al.*, 2009 |
| *Penicillium citreonigrum* | Brominated nor-eremophilane **3-46** Citreopenin **3-47** | Cytotoxic activity | Yuan *et al.*, 2014, 2015 |
| *Penicillium copticola* PSURSPG138 | Penicilleremophilane A **1-15** | Antimalarial activity | Daengrot *et al.*, 2015 |
| *Penicillium oblatum* | PF1092 A **3-48**, B **3-49**, C **3-50** | Inhibition of progesterone binding | Tabata *et al.*, 1997 Kurihara *et al.*, 1997 Tatsuta *et al.*, 1997 |
| *Penicillium roquefortii* | PR-toxin **1-5** | Inhibitor of the RNA polymerase | Moule *et al.*, 1976 |
| *Penicillium* sp. | Petasol **1-4** | Inhibitor of the regulatory protein HIV-1 | Jayasuriya *et al.*, 2005 |
| *Penicillium* sp. PR19 N-1 | Eremophilanes **3-39**, **3-40**, trinoreremo-philane **3-41** | Cytotoxic activity | Lin *et al.*, 2014 Wu *et al.*, 2013 |
| *Penicillium* sp. SS080624SCf1 | JBIR-28 **3-45** Phomenone **1-2** Sporogen AO-1 **1-3** | Cytotoxic activity | Motohohashi *et al.*, 2009 |


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| | | | |
|---|---|---|---|
| *Periconia* sp. F-31 | Periconianones A, B, D, G, K **2-22** - **2-26** | Anti-inflammatory activity | Zhang *et al.*, 2014; Liu *et al.*, 2016; |
| *Perinconia byssoides* OUPS-N133 | Peribysin A, B, C, I, D, H, E, F, G **3-30** - **3-38** | Cell adhesion inhibitors | Yamada *et al.*, 2004, 2005, 2006<br>Koshino *et al.*, 2006<br>Weber *et al.*, 1995<br>Okabe *et al.*, 1994 |
| *Phoma* sp. NRRL 25697 | Phomadecalins A, B, D **1-8** - **1-10** | Antibacterial activity | Hatakeyama *et al.*, 2010 |
| *Trichothecium roseum* | Cuspidatol **2-21** | Immunosuppressive activity | Kawamura *et al.*, 2000 |
| unidentified endophytic fungus | Xylarenone A **1-6**, B **1-7** | Antibacterial activity and antitumor activity | Hu *et al.*, 2008 |
| Unidentified xylariaceous endophytic fungus YUA-026 | Eremoxylarins A **4-51**, B **4-53** | Cytotoxicity (**4-51**), antibacterial activity (**4-51**, **4-53**) and inhibitors of calcineurin without immunophilins | Efremenkova *et al.*, 2015<br>Shiono and Murayama, 2005<br>Ogasawara *et al.*, 2008 |
| *Xylaria nigripes* | Nigriterpene C **2-16** | Anti-inflammatory activity | Chang *et al.*, 2017 |
| *Xylaria persicaria* | Xylarenal A **4-63**, B **4-64** | Antagonists of neuropeptide Y | Smith *et al.*, 2002 |
| *Xylaria* sp. BCC 21097 | α-methylene-lactones **3-42** - **3-44** | Antitumor activity (**3-42** – **3-44**), antimalarial activity (**3-43**, **3-44**) and antifungal activity (**3-44**) | Isaka *et al.*, 2010 |
| *Xylaria* sp. NCY2 | 1-(xylarenoneA)xyla-riateA **4-59**<br>Xylarenone A **1-6**, acetate<br>Xylarenone B **1-7** | Antibacterial activity and antitumor activity | Hu *et al.*, 2010 |
| *Xylaria* spp.<br>*Xylaria feejeensis* 2FB-PPM08M | Integric acid **4-54** | Inhibitor of HIV-1 integrase, antimalarial activity and weak antimicrobial activity | Singh *et al.*, 1999<br>Srisapoomi *et al.*, 2015, 2015a |
| Xylariaceous fungus LL-07H239, *Xylaria* sp. BL321 | 07H239-A **4-52** | Cytotoxic and antitumor activity | Mc Donald *et al.*, 2004; Song *et al.*, 2011, 2012, 2012a; |

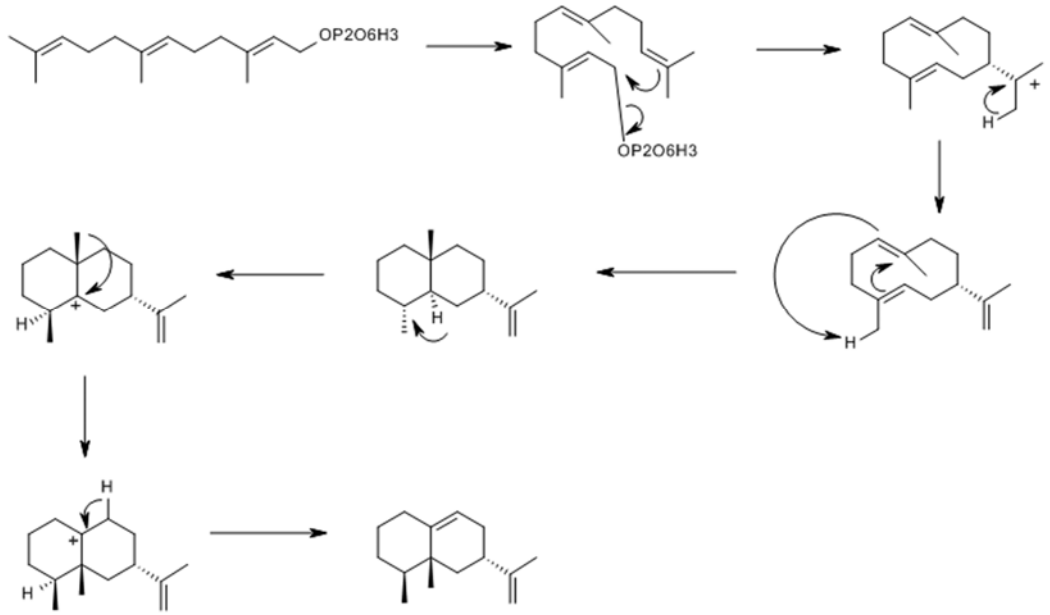


**Figure 1** Biosynthesis of eremophilane sesquiterpenes in fungi.


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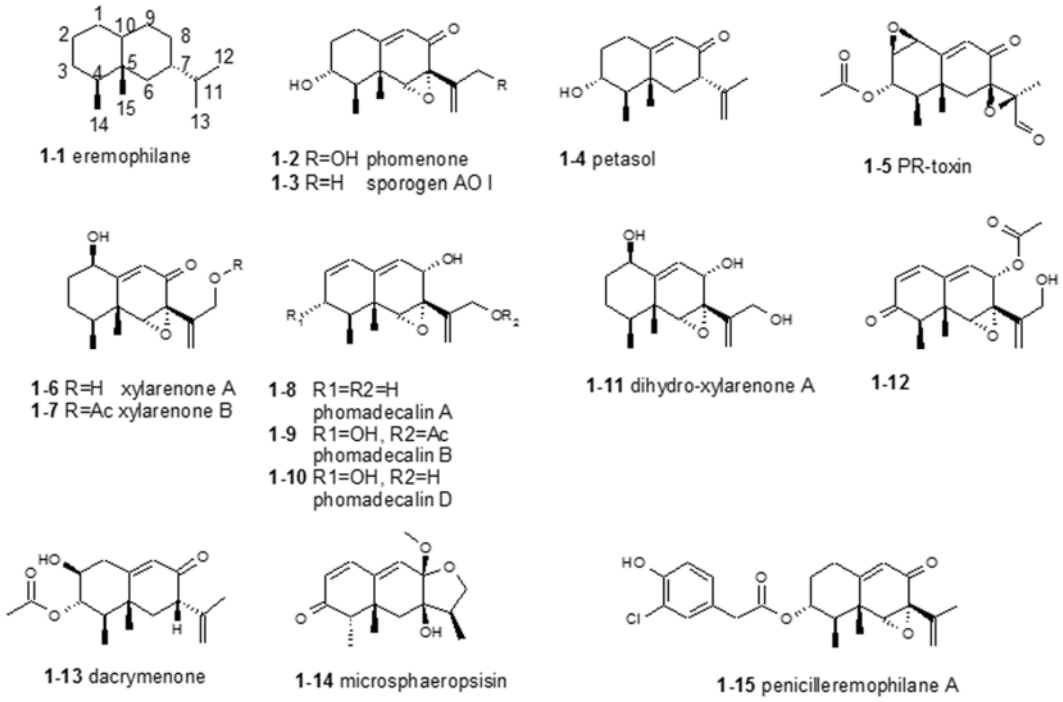


**Figure 2** Fungal eremophilanes active against bacteria and viruses.

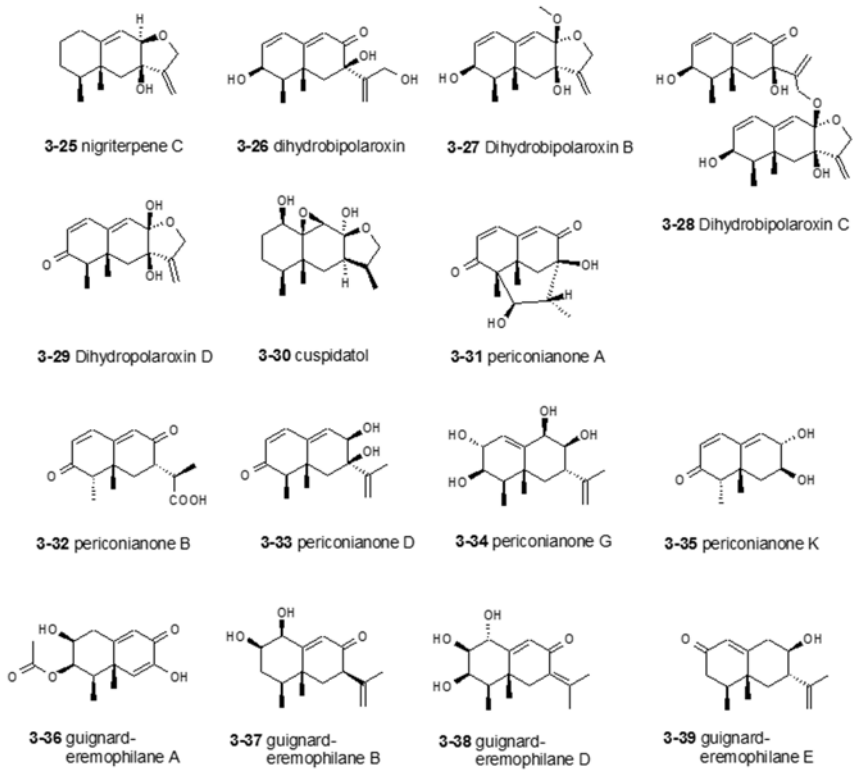


**Figure 3** Anti-inflammatory eremophilanes.


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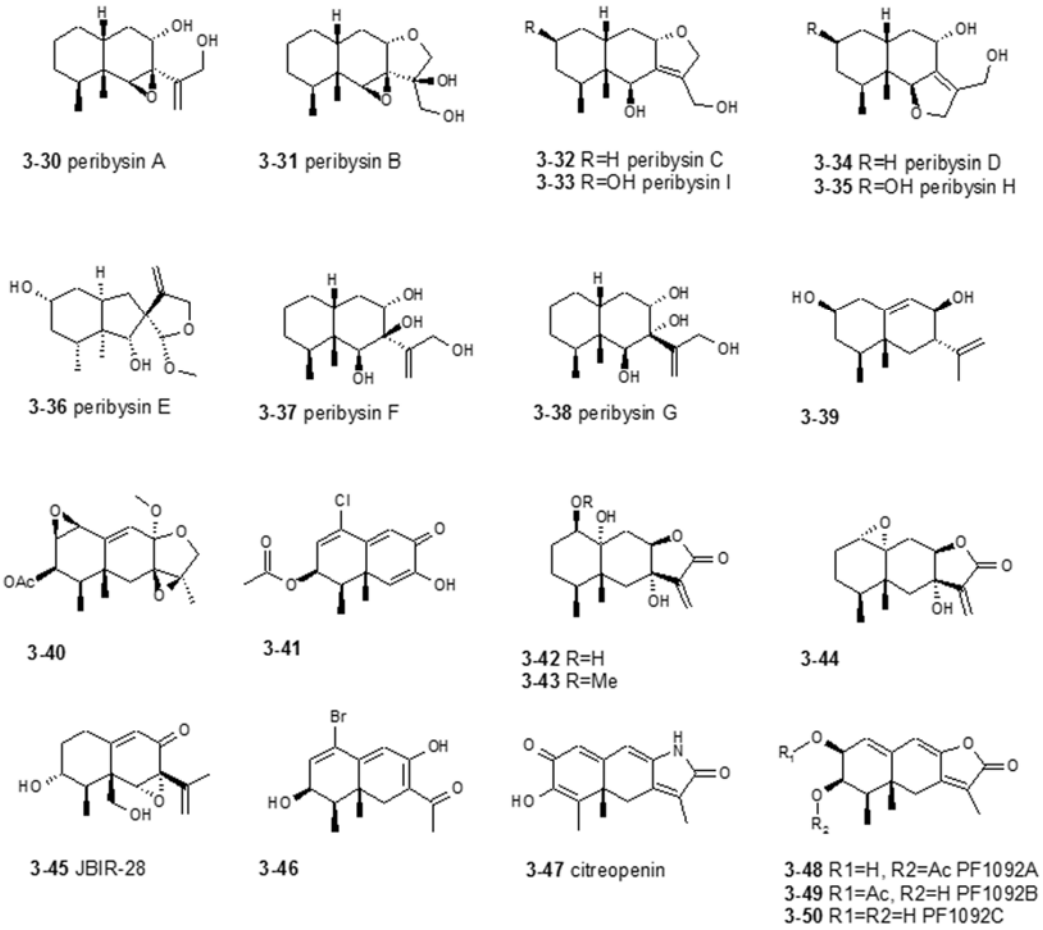


**Figure 4** Eremophilanes acting against cancer cell lines or metastases.


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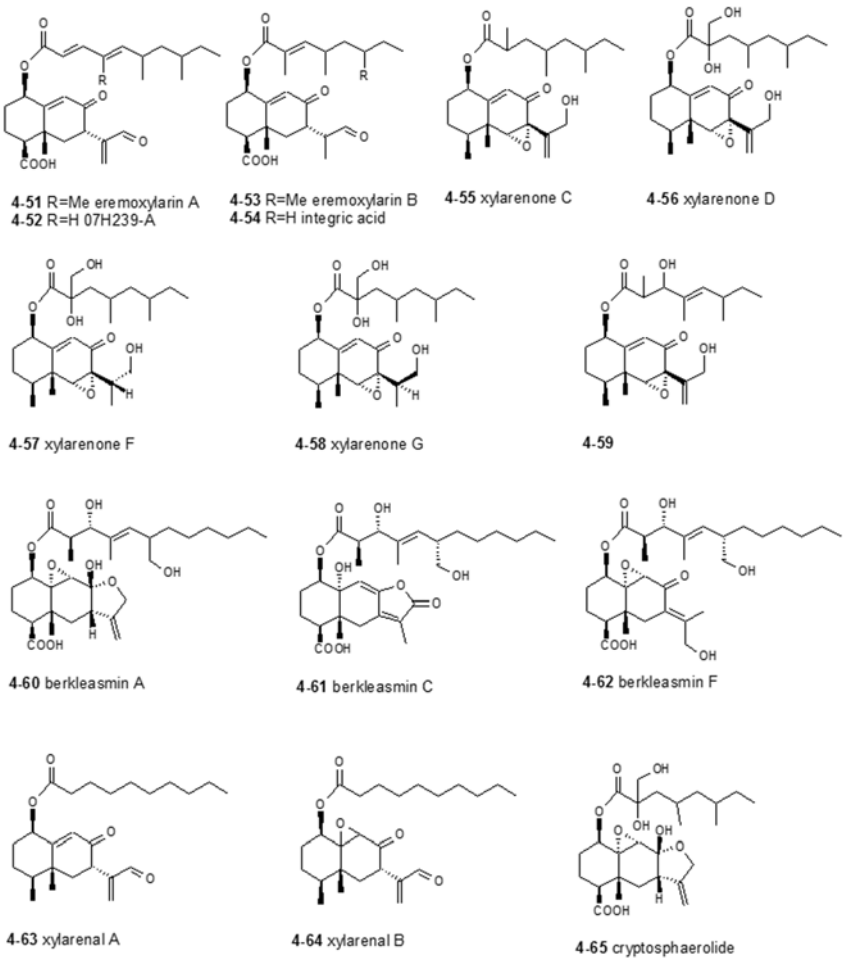


**Figure 5** Fungal eremophilanes bearing an ester moiety at C-1.

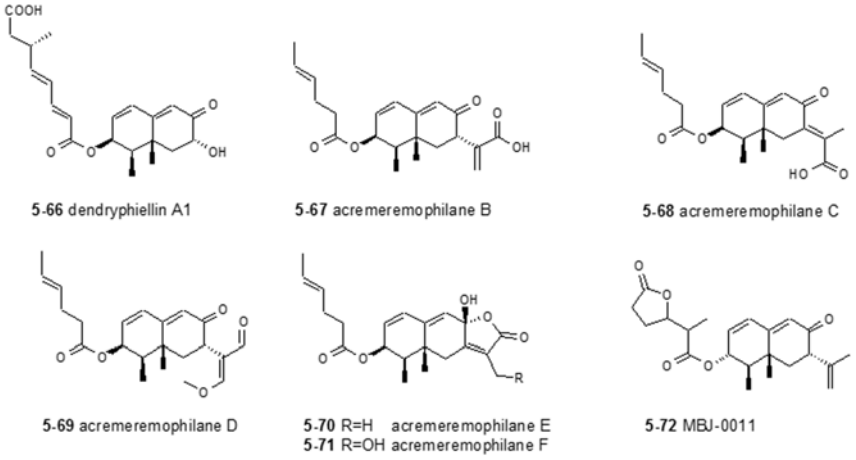


**Figure 6** Fungal eremophilanes with C-3 ester moieties.


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